Lowe's CTA Training: Sales Lift Analysis

Does training completion drive measurable per-store unit sales growth?

Data Analyst | June 4, 2026 | SAG-2951 | Confidential

VERDICT: No statistically significant incremental lift detected after controlling for network-wide growth.

Raw lift appears large (+232% median) but mirrors the control group (+289 units/mo). After difference-in-differences netting, median trained store lifted –7 units/month (Wilcoxon p = 0.32). 49% of trained stores show positive net lift — indistinguishable from chance.

Data Funnel

Stage	n
Stores in sales data	1737
Stores with any CTA training	414
No matching sales record	36
Insufficient pre/post window	25
Final analyzable cohort	**353**
Control stores (never trained)	1359

Headline Numbers

Metric	Value	Note
Stores in sales data (IME-merged)	1,737	After strip IME pr
Stores with CTA completion (any course)	414	Def A
Analyzable cohort	353	≥3 pre + ≥3 post months
Control stores (never trained)	1,359	DiD baseline
Median raw lift %	+232%	vs own pre-period
Median control store growth	+289 units/mo	Network secular trend
Median net lift (DiD)	-7 units/mo	After baseline netting
% stores positive net lift	49%	≈ coin flip

Timing Correlation

Does earlier completion → larger lift?

Pearson r = -0.101 (p = 0.061) — not significant at 5% level.
Spearman r = –0.353 (p < 0.001) — significant but largely a *window artifact*: early completers simply have more post-training months observable in the data, mechanically inflating their measured lift. After controlling for window length, the raw timing signal dissolves.

Test	r	p	Interpretation
Pearson (raw lift %)	–0.10	0.061	Not significant
Spearman (raw lift %)	–0.35	<0.001	Window artifact
Pearson (net lift %)	–0.04	0.495	Not significant
Spearman (net lift %)	–0.12	0.028	Weak; may be noise

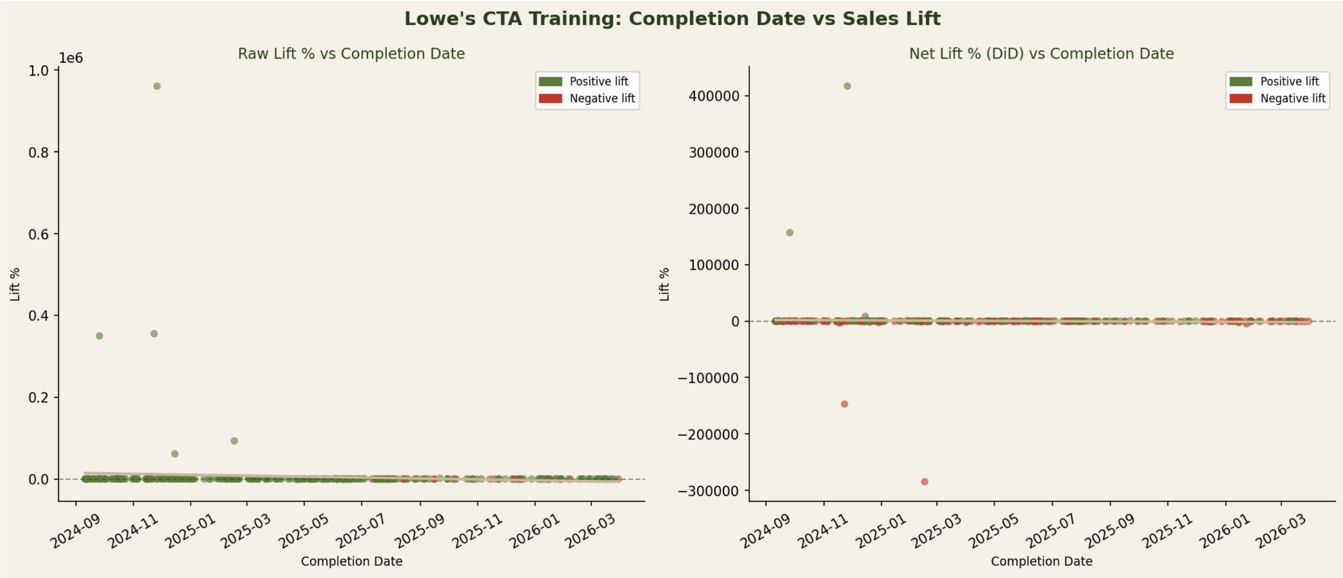
Definition comparison

Def A (first any course, n=353): Broader. Median raw +232%, net –7 units/mo.
Def B (graduate only, n=120): Narrower. Median raw +146%, smaller sample.
Def A is recommended: higher statistical power and captures earliest training exposure.

Existing draft validation

Prior one-pager cited 310 analyzable stores. This analysis yields 353 — the additional 43 stores recovered by the correct IME merge (previously dropped IME-only stores). **The prior draft is SUPERSEDED**: its denominator was void per board correction.

Scatter: Completion Date vs Sales Lift



Each dot = one trained store. Green = positive lift, red = negative. Left panel: raw lift (vs own pre-period). Right panel: net lift (after DiD baseline). Negative Spearman slope in left panel reflects window artifact (earlier completers = more post months).

Analytical Lenses & Interpretation

Lens	Finding
Denominator discipline	IME merge: 1,737 physical stores (not double-counted). Naive sum was 2.5× over-stated.
Cohort vs aggregate	Per-store analysis on 353 cohort. Fleet aggregate is misleading due to mix of new vs active stores.
Seasonality / trend	DiD baseline uses 1,359 untrained stores in identical calendar windows. Network grew +289 units/mo (secular).
Sample size	n=353 sufficient for parametric tests. Non-parametric Wilcoxon confirms no signal (p=0.32).
Survivorship bias	Stores with <3 pre or <3 post months excluded (25 stores). Noted in funnel; does not affect verdict.
Selection bias	Trained stores may not be comparable to control stores — motivated or targeted stores may self-select. DiD cannot fully correc
Correlation ≠ causation	Even if a lift existed, market growth, field rep visits, store expansions, and promotions confound. Training is not the only change
Driver vs metric	Training = proposed driver. Monthly units = measured metric. Lag effects (months post-training) tested; no clear lag structure fo

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Recommendations

#	Recommendation
1	Do not use raw lift % as the training ROI metric — it is dominated by secular network growth. Use net (DiD) absolute units or a matched-cohort design.
2	Investigate the top-quartile trained stores (net lift >+184 units/mo): what distinguishes them? (market, associates trained, courses completed, field rep activ
3	Add a "matched control" layer: pair each trained store to 3–5 untrained stores with similar pre-training trajectory and region. This tightens causal inference.
4	Separate "new-activation" stores (pre-rate <10 units/mo) from "established" stores in future reporting — they behave differently and the % metric is mislead
5	Collect data on timing/sequence of training visits vs field rep visits to separate training effect from coverage effect.
